CS3IP Project Definition Form

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Course: CS3IP (BSc Computer Science)

Project title: Evaluating Temporal Knowledge-Graph Retrieval for Evidence-Grounded Question Answering on Crisis Events

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Project overview:
This project is set out to evaluate the accuracy and relevance of responses given by AI when they are provided with linked data points and an evidence-grounded question on the topic. Evaluation for such a system is needed more and more the further people reply on AI for reliable information; knowing if these responses can be made more accurate (lowering hallucination and variance rate, etc.) should provide peace of mind and a better understanding of how to configure AI models when it comes to giving responses for critical conversations.

Project deliverable:
A chatbot that will detect disaster-related messages and use web-scrapping/API's to build a dataset that can be vectorised using embedding models to create a "temporal knowledge graph" for answering evidence-based questions. This system should be evaluated to determine the level of relevance to current chat (is response grounded in the same context as current chat) and accuracy to the knowledge graph.
UX work will be added if time allows, but building the base system and evaluation of that system come first.

What is interesting & useful about this project? Can you make a business case for it?
This project will allow us to understand the degree to which AI/LLM output is reliable enough to use for data analysis of critical events, when such technology is given a trusted data-set with which it can retrieve data.

Business Case: Results of my research in this topic will show reliability of AI chatbots for data analysis of critical events, their reliability and accuracy to source material. If shown to be reliable and accurate then this system used for testing can act as a skeleton for future architecture when developing AI chatbots that would be used for similar situations; if results are negative then we can conclude that the use of "Temporal Knowledge Graphs" for AI data retrieval is not sufficient for giving responses on critical event datasets.

Taking the system further I could add features like;
- Creating embeddings from given .xml file
- End-User configurability of system
- UI/X design

Outline Project Timetable:

Gantt Chart built within GitHub Project View (Public view available).
See: https://github.com/users/ProfProgram/projects/3/views/1